

Development of an Experimental Measurements Database for granular materials

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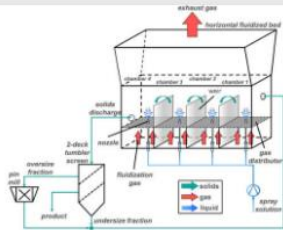


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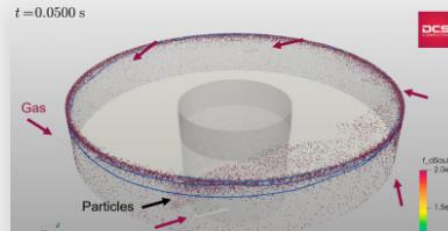
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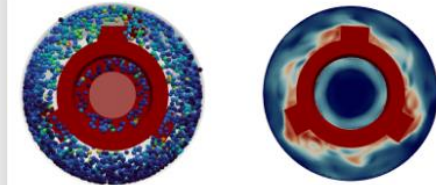
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Experimental Measurements Database v1

[Experimental Database | TUSAIL](#)

Sr. No	ESR	Stage	Name/type of experiment	Material Name	Size (any of d50, d32, d90, d10)	Shape
1	1		Powder topology - Material state	Maltodextrin DE 20 (Avebe MD 20 P)	d50 ~ 180µm	Aspherical
2	1		Agglomeration/Fluidization	Maltodextrin DE 20 (Avebe MD 20 P)	d50 ~ 180µm	Aspherical
3	1		Glass transition temperature	Maltodextrin DE 20 (Avebe MD 20 P)	d50 ~ 180µm	Aspherical
4	1		Liquid penetration kinetics	Maltodextrin DE 20 (Avebe MD 20 P)	d50 ~ 180µm	Aspherical
5	2		Tableting	Dried glucose syrup DE29	d50= {99µm, 254µm}	Aspherical
6	2		Particle size distribution	Dried glucose syrup DE29	d50= {99µm, 254µm}	Aspherical
7	2		Diametral crushing	Dried glucose syrup DE29	d50= {99µm, 254µm}	Aspherical
8	2		3 point bending test	Dried glucose syrup DE29	d50= {99µm, 254µm}	Aspherical
9	2		Tablets surface topology	Dried glucose syrup DE29	d50= {99µm, 254µm}	Aspherical
10	2		Mercury intrusion	Dried glucose syrup DE29	d50= {99µm, 254µm}	Aspherical
11	2		Particle surface topology	Dried glucose syrup DE29	d50= {99µm, 254µm}	Aspherical
12	3		Jet mill (BASF)	CaCO3	d100 ~ 150µm	Spherical
13	3		Breakage characterization	CaCO3	d100 ~ 150µm	Spherical
14	4		Wet Grinding (Stirred Media Mill): material	ESKAL 30 (CaCO3)	d50 ≈ 30µm	-
15	4		Wet Grinding (Stirred Media Mill): Medium	Zirconia Beads (ZrO2)	d50 ≈ 315µm, 730µm	Spherical

Experimental Measurements Database v2

[TUSAIL EM Db \(jpmorr.eu.pythonanywhere.com\)](http://TUSAIL EM Db (jpmorr.eu.pythonanywhere.com))

Materials Database

The searchable materials database is shown below. New materials can be added to the database.

Submit New Material Metadata

View Equipment Metadata

View Experimental Records

Show entries

Search:

ID	Name	Product Name	Classification	Nature	Shape Description	Size Distribution	Size D50	Size Dmin	Size Dmax
1	Limestone	Eskal-500	Powder	Cohesive	Spherical	Poly-disperse	4.9	1.44	14.1
2	Limestone	Eskal-300	Powder	Cohesive	Spherical	Poly-disperse	3.4	1.0	9.6
3	Limestone	Eskal-10	Powder	Cohesive	Spherical	Poly-disperse	11.6	2.64	24.0
4	Limestone	Eskal-20	Powder	Cohesive	Spherical	Poly-disperse	22.0	6.89	38.75

Experimental Measurements Database v3

DRS Ambassador internship project – looks identical to v2, but..

Materials Database

The searchable materials database is shown below. New materials can be added to the database.

**Submit New Material
Metadata**

**View
Equipment Metadata**

**View
Experimental Records**

Show entries

Search:

ID	Name	Product Name	Classification	Nature	Shape Description	Size Distribution	Size D50	Size Dmin	Size Dmax
1	Test Material	Test Product	Ceramic	Amorphous	Angular	Bi-disperse			
2	Another Material	Testy thingy	Boulder	Cohesionless	Irregular	Mono-disperse	22	1	25
3	Limestone Powder	Eskal-500	Powder	Cohesive	Spherical	Poly-disperse	4.9	1.44	14.1

Changes: backend database

V2: SQLite

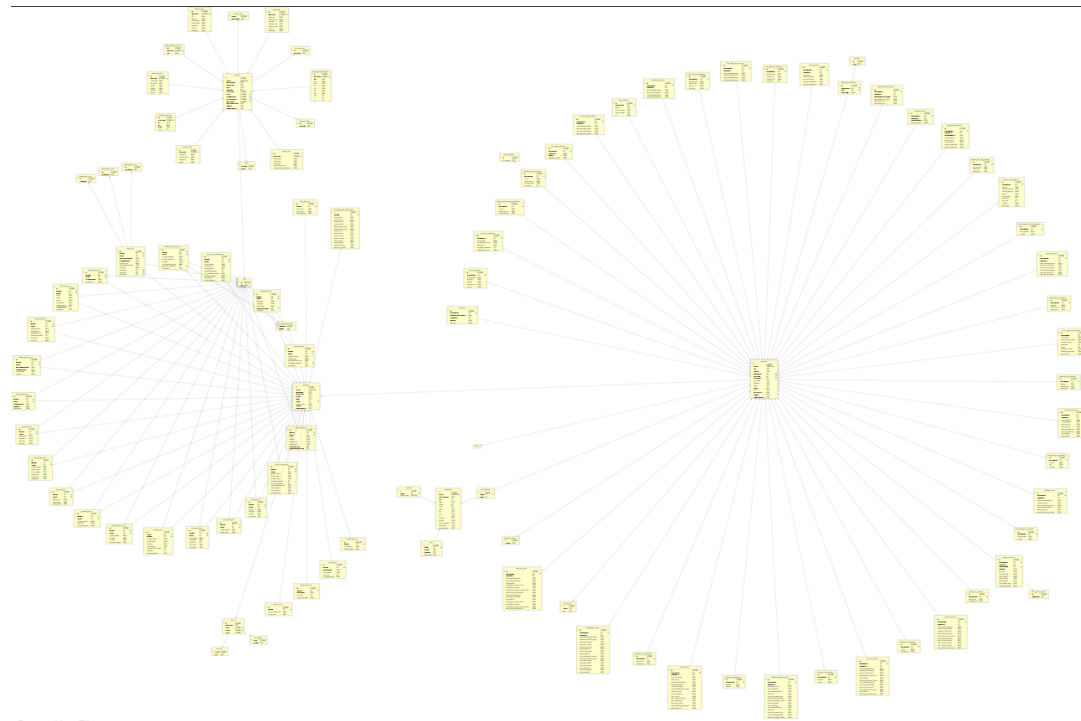


V3: PostgreSQL



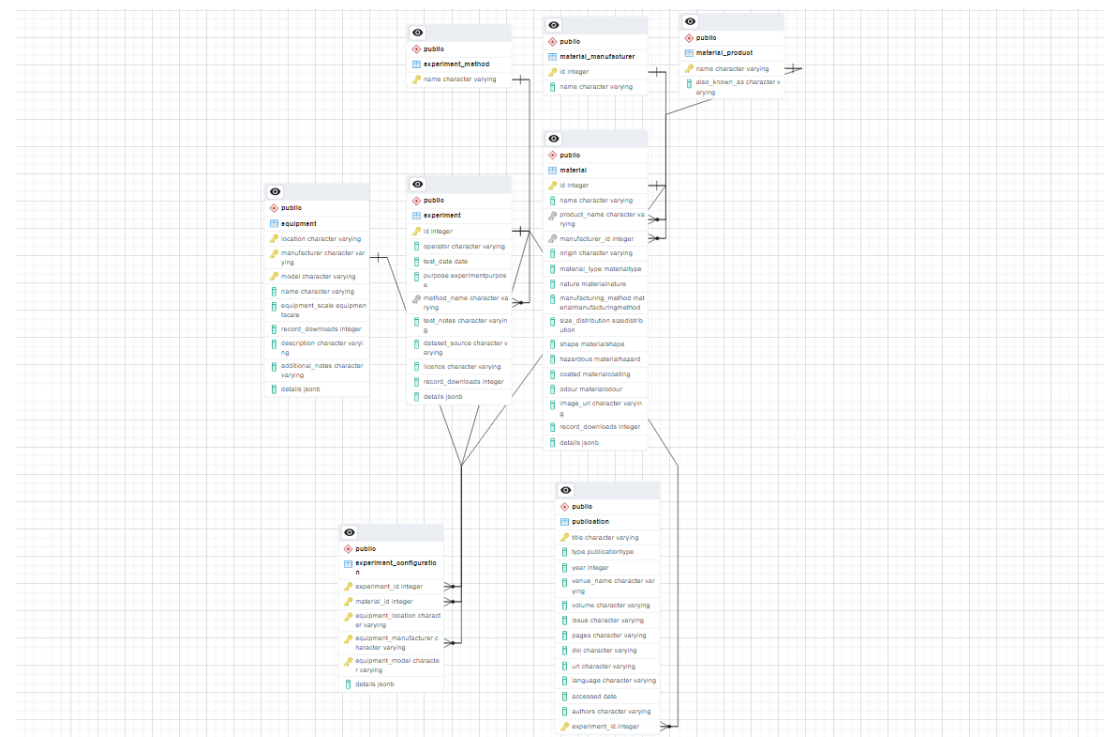
Changes: database tables

V2: ~50 tables



Downloaded by yFiles

V3: 8 tables



Changes: adding new materials/equipment/experiment types

V2: 1000s lines of duplicated code

```

irements.txt material.py equipment_view.html (f1c0322) ← equipment_view.html (dffe813)
C: > Users > Vidminas > GitHub > tusail-emdb > web > templates > equipment > equipment_view.html > ...
2493-         </tbody>
2494-     </table>
2495- </div>
2496- <div class="card-footer">
2497- </div>
2498- </div>
2499- </div>
2500- </div>
2501-iv>
2502-v class="36 equiptype_selection">
2503- <div class="row mb-3">
2504-     <div class="col-sm-12 col-md-6 col-lg-6">
2505-         <div class="card">
2506-             <h5 class="card-header">Geometric Pro
2507-             <div class="card-body">
2508-                 <table class="table table_view">
2509-                     <tbody>
2510-                         <tr>
2511-                             <td style="vertical-a
2512-                             <td style="vertical-a
2513-                         </tr>
2514-                         <tr>
2515-                             <td style="vertical-a
2516-                             <td style="vertical-a
2517-                         </tr>
2518-                         <tr>

```

V3: one JSON file per entity

```

1  {
2      "$id": "equipment/imaging_high_speed.v1-0-0.json",
3      "$schema": "https://raw.githubusercontent.com/Vidminas/python-jjsonschema-
4
5      "type": "object",
6      "properties": {
7          "test_category": { "const": "Imaging" },
8          "test_subcategory": { "const": "High Speed" },
9          "class": { "const": "Dynamic" },
10         "rating": { "const": 3 },
11         "optical_properties": { "$ref": "#/$defs/optical_properties" },
12         "measurement_parameters": { "$ref": "#/$defs/measurement_parameters" }
13     },
14     "required": ["optical_properties", "measurement_parameters"],
15
16     "$defs": {
17         "optical_properties": {
18             "type": "object",
19             "properties": {
20                 "colour": { "enum": ["Black & White", "Colour"] },
21                 "sensor": { "type": "string" },
22                 "lens_mount": { "type": "string" },
23                 "lens": { "type": "string" },
24                 "operating_temperature_range": { "type": "string" }
25             },

```

Changes: units

V2: implicit/fixed units

Physical Properties	
True Density:	2800.0
Envelope Density:	2650.0
Specific Gravity:	
Particle Porosity:	

Mechanical Properties	
Shear Modulus:	
Young's Modulus:	
Poissons Ratio:	
Hardness:	3.0
Toughness:	
Ductility:	

V3: convertible units

physical_properties		
colour	N/A	
coating material	N/A	
refractive_index	N/A	-
true_density	N/A	kg/m^3
envelope_density	N/A	kg/m^3
specific_gravity	N/A	None
particle_porosity	N/A	None

mechanical_properties		
shear_modulus	N/A	Pa
youngs_modulus	N/A	Pa
poissons_ratio	N/A	None
mineral_hardness	N/A	moHS
hardness	N/A	N/mm^2
toughness	N/A	Pa
ductility	N/A	None

Changes: experiments with multiple materials/equipment

V2: one material/equipment per experiment

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Home / Experiments Database

Experiments Database

The searchable experimental records database is shown below. New experiments can be added to the database.

[View Material Metadata](#) [View Equipment Metadata](#) [Submit New Experimental Record](#)

Show entries Search:

ID	Test Date	User	Material	Equipment	Location	Category	Phase	Purpose	Quality	Dataset	Lic
1	2012-06-12	J.P. Morrissey	Limestone	Edinburgh Powder Tester	University of Edinburgh, Edinburgh, UK	Uniaxial Test	Dry	Calibration	3.7	gitlab.com/jpmorr	CC 4.0

V3: multiple materials/equipment per experiment

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Experiments Database

The searchable experimental records database is shown below. New experiments can be added to the database.

[View Material Metadata](#) [View Equipment Metadata](#) [Submit New Experimental Record](#)

Show entries Search:

ID	Operator	Test Date	Materials and Equipment	Purpose	Dataset	Licence
1	Vid	2000-12-20	Test Material with Edinburgh pilot shear tester Test Material with Edinburgh big shear box	Verification	fake	ignored
2	Vid	2000-04-20	Another Material with Edinburgh huge rotating drum tester	Calibration	classified	ignored

Minor additions: images, bounds validation

Material and equipment images

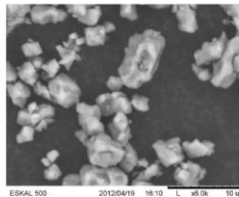
[Home](#) / [Materials](#) / View Material

Material ID: 3 - Eskal-500

Download Record
Metadata

Download metadata for this
material.

Download json



- Description

Manufacturer:	KSL Staubtechnik GmbH
Product Code:	PARDEM Reference Solid
Origin:	None
Material Type:	Powder

Size Distribution

Measurement Unit:	mm			
D50:	4.9	D10:	1.75	
Dmin:	1.44	D30:	3.4	
Dmax:	14.1	D60:	6.0	

Relative min/max bounds checking

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- ['size_distribution']: 'Dmin'=5.0 is greater than the maximum 'D50'=3.0!

OK

Outputs summary

Public library pull requests / discussions:

[Handle Enum columns backed by Python Enums by Vidminas · Pull Request #164 · kvesteri/wtforms-alchemy \(github.com\)](#)

[How to apply custom validator to schemas that specify \\$schema? · Issue #1133 · python-jsonschema/jsonschema \(github.com\)](#)

[unevaluatedProperties error gets raised before more relevant validation errors · Issue #1132 · python-jsonschema/jsonschema \(github.com\)](#)

git diff --stat main sqlalchemy

- 176 files changed, 7083 insertions(+), 21199 deletions(-)

What's next for the TUSAIL EMDB?



Consultations + evaluations with TUSAIL PhD students / post-docs



New Horizon Europe grant application in October to expand the EMDB